Assignment 1-EPPS 6323

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Brainstorming Ideas for Knowledge Mining projects from an AI perspective

Knowledge mining in Artificial Intelligence (AI) is an exploratory area that involves extracting valuable inferences from large amounts of structured and unstructured data. Some brainstormed ideas across different domains are noted below:

Scientific Research & Academia

Automated Literature Review: AI tools that go through and summarize academic papers, extracting crucial findings and trends.

Correlating Knowledge from different fields : AI that correlates research findings across different fields (e.g., AI mining chemistry and biology papers for interdisciplinary comparison).

Plagiarism Detection: AI models that compare research content against existing databases to detect novelty.

Environmental & Climate Data Mining

Weather & Climate predictions: Mining satellite images, ocean data and weather patterns for climate insights.

Renewable Energy Optimization: AI that mines solar, wind, and hydro-energy datasets to improve efficiency.

Biodiversity & Conservation Insights: AI that processes ecological data to analyze endangered species and forest degradation.

Government & Policy decisions

Policy Impact Analysis: AI tools that mine government policies, analyze public reactions and predict societal impacts.

Urban planning: AI that extracts insights from urban traffic, pollution and citizen complaints to improve city planning.

AI for Crime Prediction & Prevention: Mining law enforcement data, social media and public reports to analyze and prevent crime.

Some of the Geo-Information and Science (GIS) based ideas looked into are as follows:

Weather Data Scraper with GIS Visualization

Scrape real-time weather data (temperature, humidity, wind speed) from APIs like OpenWeatherMap and plot it on interactive maps.

Tools: Python (Requests, BeautifulSoup), Leaflet.js, Folium.

Urban Traffic Data Scraper

Collect traffic data (like congestion levels) from websites such as Google Maps Traffic API and visualize hotspots in a city.

Tools: Selenium, OpenStreetMap, Folium.

Wildfire Incident Tracker

Collect wildfire data from NASA's Fire Information for Resource Management System (FIRMS) and map active fires globally.

Tools: Python, Leaflet, Satellite Imagery APIs.

AI-based drone data projects from a knowledge mining perspective that combine drone data with artificial intelligence techniques:

3D Terrain Mapping

Idea: Generate 3D models of terrains using drone-captured images. Apply AI to analyze land use, erosion, and urban expansion.

Knowledge Mining: Understand topographical changes, predict landslides, and plan infrastructure.

Tech Stack: Photogrammetry, 3D Reconstruction (OpenDroneMap), AI.

Land Cover Classification

Idea: Classify land types (forests, water bodies, urban areas) from drone images using AI models.

Knowledge Mining: Track land use changes, deforestation rates, and urban sprawl.

Tech Stack: Remote Sensing, Image Classification, GIS Tools.

Drone Flight Path Optimization

Idea: Analyze historical flight data to optimize future drone missions for battery efficiency and coverage.

Knowledge Mining: Predict optimal flight paths based on environmental conditions and past performance.

Tech Stack: Time-Series Analysis, AI Path Planning Algorithms.

Narrowed down Project Idea

Drone Permitted and No- fly zone Identification

Interactive Map for No-Fly Zones

- Visualize restricted areas like airports, military zones, government buildings, and national parks.
- Show dynamic restrictions (e.g., temporary no-fly zones during events).
- Highlight areas where drone flights are permitted possibly with altitude and time restrictions.

- Suggest optimized flight paths avoiding restricted zones.

Data Sources

- AirMap API: Provides airspace regulations, no-fly zones, and real-time restrictions.
- OpenSky Network: Flight tracking data for identifying high-traffic airspace.
- FAA UAS Data: No-fly zone data for the United States.
- OpenStreetMap (OSM): Base maps and additional geospatial data.

Steps to implement:

Data Collection and Integration

- Fetch no-fly and allowed zone data via APIs.
- Convert data into GeoJSON format for map rendering.

Map Visualization

Use **Leaflet.js** or **Mapbox** to display zones with color symbol coding:

- Red for no-fly zones
- Green for allowed zones
- Yellow for restricted areas (with conditions)

AI-Based Route Planning

- Implement AI algorithms to suggest optimal drone flight paths while avoiding restricted areas.

Keeping track of Legalities:

- Data Compliance: Ensure compliance with regulations put forth by aviation authorities.
- Privacy Concerns: Take care not to handle sensitive flight data without appropriate permissions.

References

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